

Description and choice data for the domain “Sentences”

Description of the choice domain 14, Sentences

The prompt question and the universe of five response options in the choice domain **Sentences** are as follows. The labels *a*, *b*, *c*, *d* and *e* were not displayed during the experiment and are indicated here to allow cross-referencing with data tables and visualizations below and results in the paper.

This domain consists of sentences that have been used in experiments of linguistic judgement. They are, respectively sentences 7j, 7h, 7p, 7a and 7e in Bard et al. (1996). Figure 1 of that paper shows acceptability scores for these and other sentences given by two individual linguists, an acceptability score aggregating the scores of four linguists and an acceptability score aggregating the scores of four “naive respondents”, all undergraduate anatomy students. There is broad, but not perfect, agreement in terms of order, and in the following list, they are in decreasing order of acceptability according to the measure aggregating the judgements of four linguists.

Which one of the following sentences do you find the most grammatically acceptable?

- Who did Bill buy the car to please?
- This is a book which reading would be fun.
- Where did Bill buy the car to drive?
- Which man do you wonder when to meet?
- With which pen do you wonder what to write?

The following figure is a screenshot from the actual experiment, with one of the 26 possible menus for this domain.

Sentences

Which one of the following sentences do you find most grammatically acceptable?

☐ This is a book which reading would be fun.

☐ Which man do you wonder when to meet?

☐ With which pen do you wonder what to write?

☐ Who did Bill buy the car to please?

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Figure 1: Screenshot for domain Sentences

Menu A	Choice counts					Choice proportions				
	$N_A(a)$	$N_A(b)$	$N_A(c)$	$N_A(d)$	$N_A(e)$	$\hat{P}_A(a)$	$\hat{P}_A(b)$	$\hat{P}_A(c)$	$\hat{P}_A(d)$	$\hat{P}_A(e)$
$\{a, b\}$	22	19	-	-	-	0.537	0.463	-	-	-
$\{a, c\}$	13	-	27	-	-	0.325	-	0.675	-	-
$\{b, c\}$	-	16	24	-	-	-	0.400	0.600	-	-
$\{a, b, c\}$	20	3	17	-	-	0.500	0.075	0.425	-	-
$\{a, d\}$	28	-	-	13	-	0.683	-	-	0.317	-
$\{b, d\}$	-	30	-	10	-	-	0.750	-	0.250	-
$\{a, b, d\}$	21	15	-	4	-	0.525	0.375	-	0.100	-
$\{c, d\}$	-	-	30	10	-	-	-	0.750	0.250	-
$\{a, c, d\}$	16	-	19	5	-	0.400	-	0.475	0.125	-
$\{b, c, d\}$	-	12	22	6	-	-	0.300	0.550	0.150	-
$\{a, b, c, d\}$	10	8	18	4	-	0.250	0.200	0.450	0.100	-
$\{a, e\}$	19	-	-	-	21	0.475	-	-	-	0.525
$\{b, e\}$	-	19	-	-	21	-	0.475	-	-	0.525
$\{a, b, e\}$	17	12	-	-	11	0.425	0.300	-	-	0.275
$\{c, e\}$	-	-	26	-	14	-	-	0.650	-	0.350
$\{a, c, e\}$	11	-	17	-	12	0.275	-	0.425	-	0.300
$\{b, c, e\}$	-	8	24	-	8	-	0.200	0.600	-	0.200
$\{a, b, c, e\}$	8	9	15	-	8	0.200	0.225	0.375	-	0.200
$\{d, e\}$	-	-	-	6	34	-	-	-	0.150	0.850
$\{a, d, e\}$	20	-	-	2	18	0.500	-	-	0.050	0.450
$\{b, d, e\}$	-	18	-	4	18	-	0.450	-	0.100	0.450
$\{a, b, d, e\}$	13	10	-	2	15	0.325	0.250	-	0.050	0.375
$\{c, d, e\}$	-	-	28	1	11	-	-	0.700	0.025	0.275
$\{a, c, d, e\}$	13	-	14	5	8	0.325	-	0.350	0.125	0.200
$\{b, c, d, e\}$	-	10	17	3	10	-	0.250	0.425	0.075	0.250
$\{a, b, c, d, e\}$	15	7	16	0	2	0.375	0.175	0.400	0.000	0.050

Table 1: Observed choice counts and proportions.

Choice data for domain 14, Sentences

Table 1 shows choice counts and choice proportions for this choice domain. For each menu A and each object $x \in \{a, b, c, d, e\}$, $N_A(x)$ is the number of participants who chose object x from menu A and $\hat{P}_A(x)$ is the corresponding proportion of participants who chose x from A . When $x \notin A$, a dash is displayed.

The following figure displays choice proportions for all doubleton and tripleton menus in Barycentric coordinates. See a full description of this graphical representation in the paper. Each panel shows choice proportions for all doubleton and tripleton menus of a different tripleton subset of $\{a, b, c, d, e\}$. The downward-pointed (blue) triangle shows the set of ternary choice proportions that are compatible with regularity and the three binary choice proportions, on the corresponding tripleton. The upward-pointed (red) triangle shows the set of ternary choice proportions compatible with the multiplicative inequality and the three binary choice proportions.

References

Ellen Gurman Bard, Dan Robertson, and Antonella Sorace. Magnitude estimation of linguistic acceptability. *Language*, 72(1):32–68, 1996.